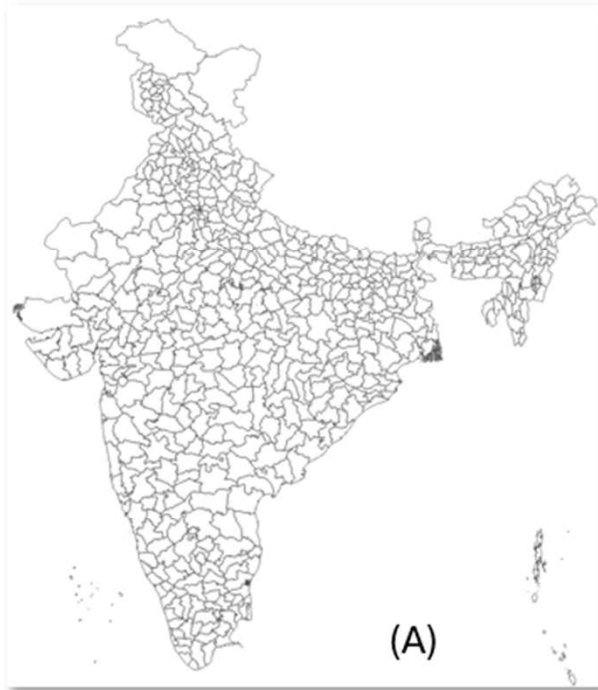
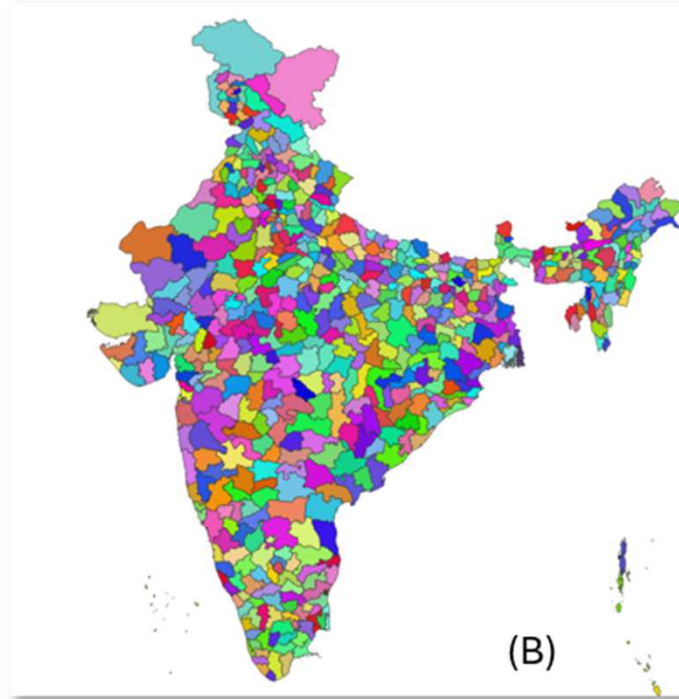


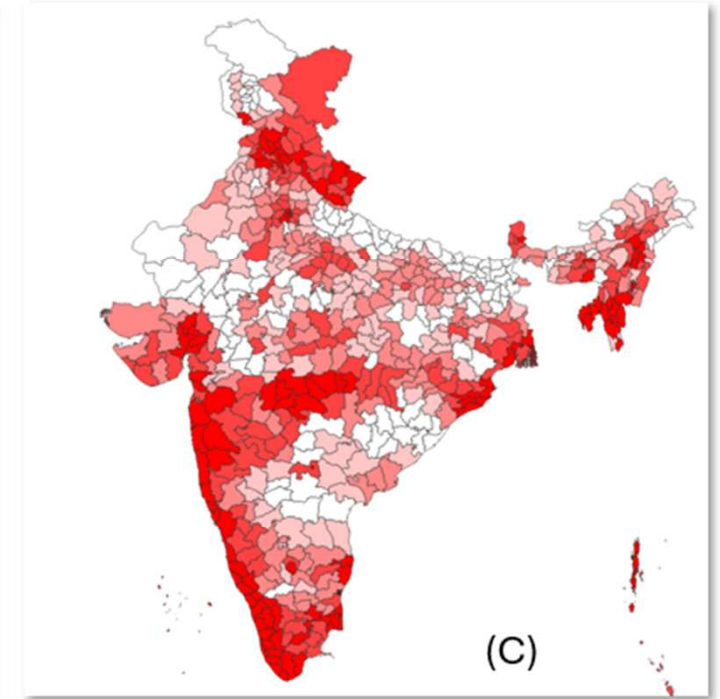
Attributes Classification Methods



Map showing all districts in India



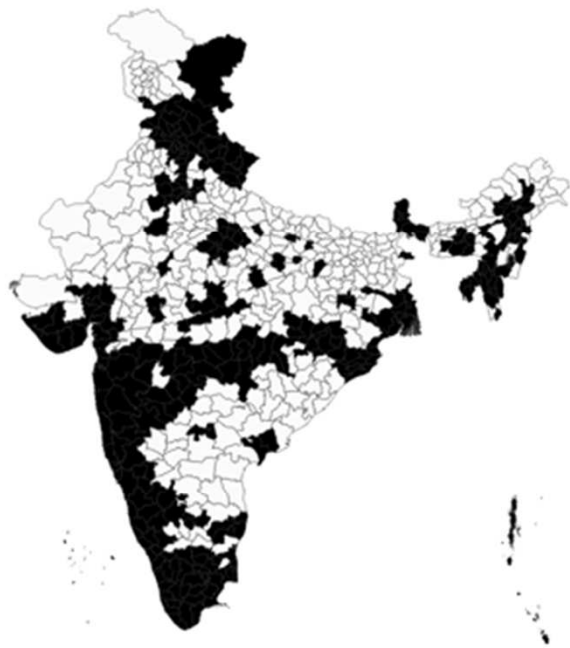
Districts randomly color-coded



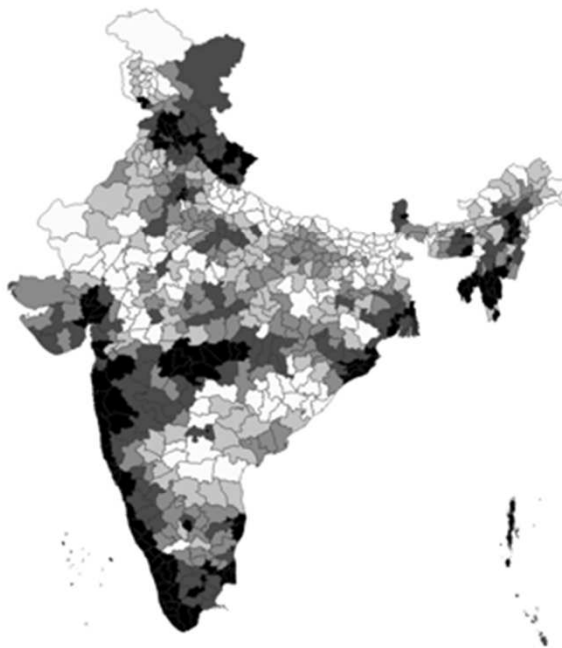
Districts classified based on literacy rate

Which map is more informative and why?

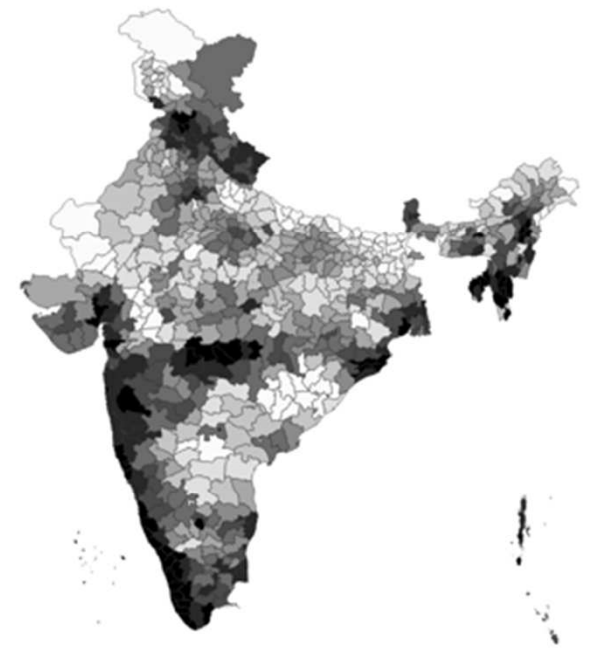
Statistical approach to Attribute Generalization



2 ranges



5 ranges

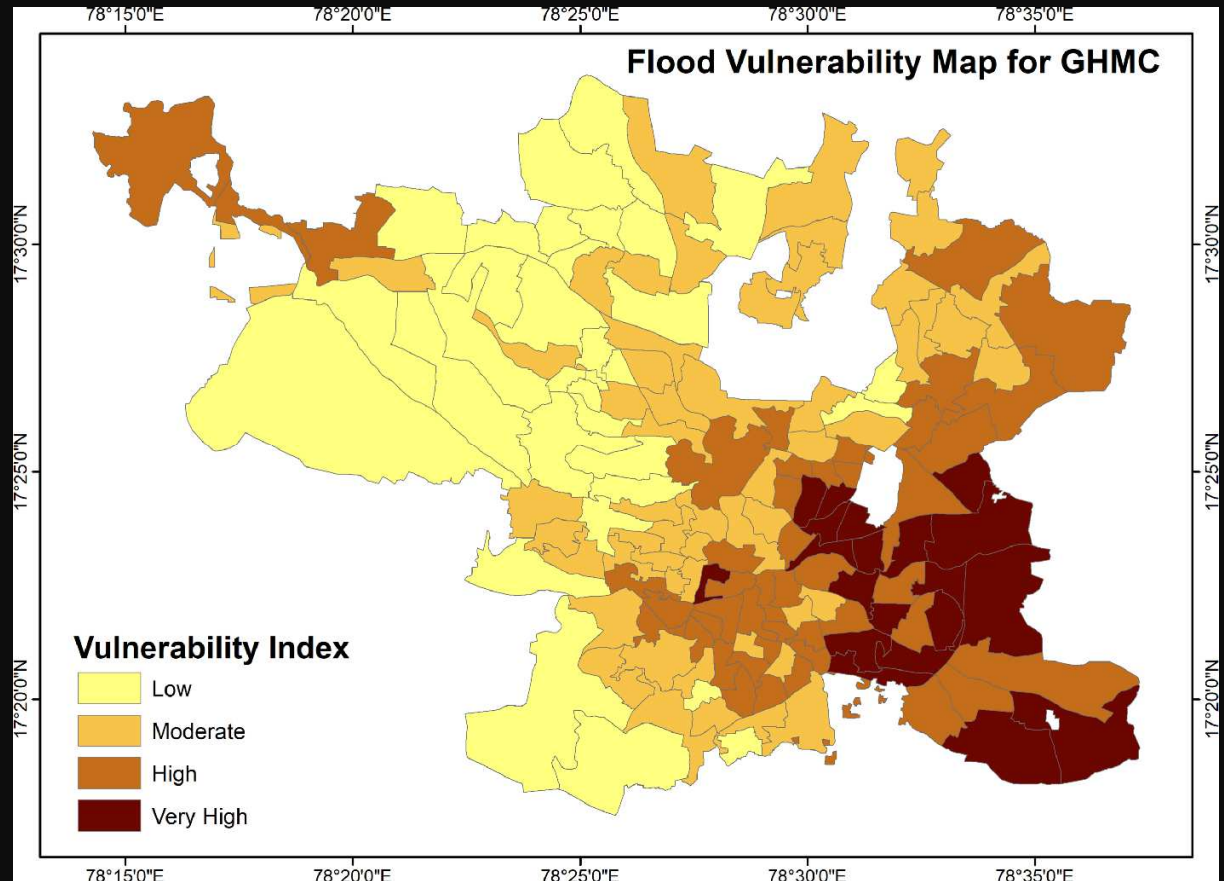


9 ranges

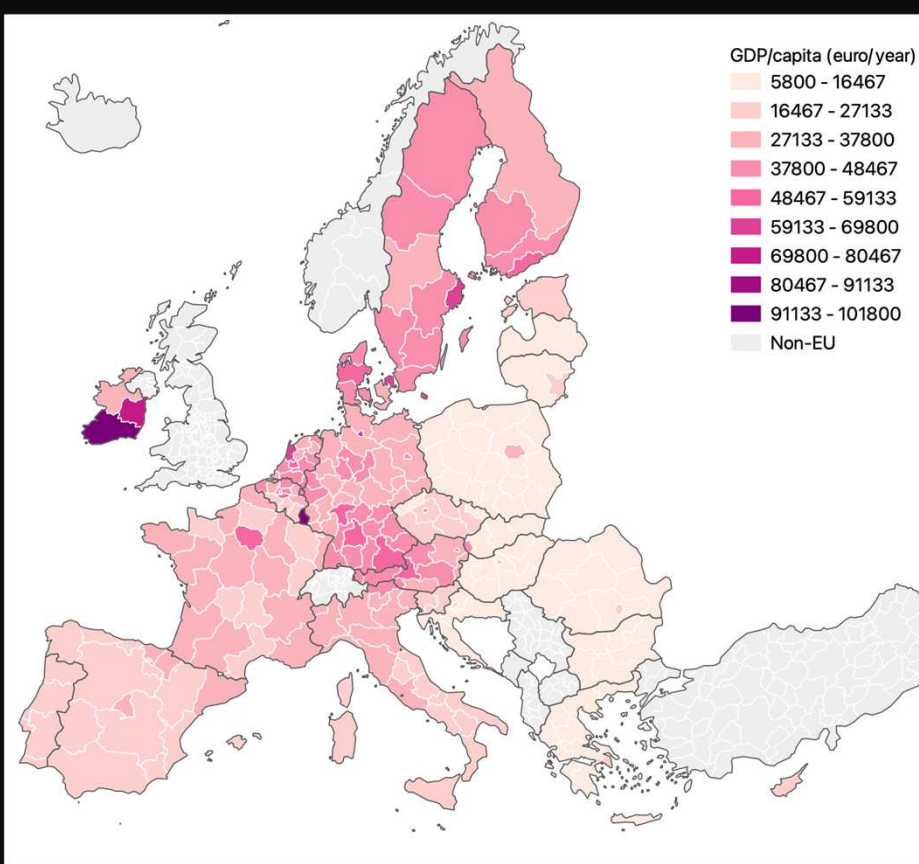
Choropleth maps

a thematic map in which polygon features are shaded or colored according to the value of an attribute.

Literacy rate (%),
Population density
(people/km²),
Forest cover (%)
Rainfall (mm),
Unemployment rate (%)



A choropleth map requires **two decisions**:
Which attribute will be mapped?
How will the values be classified?



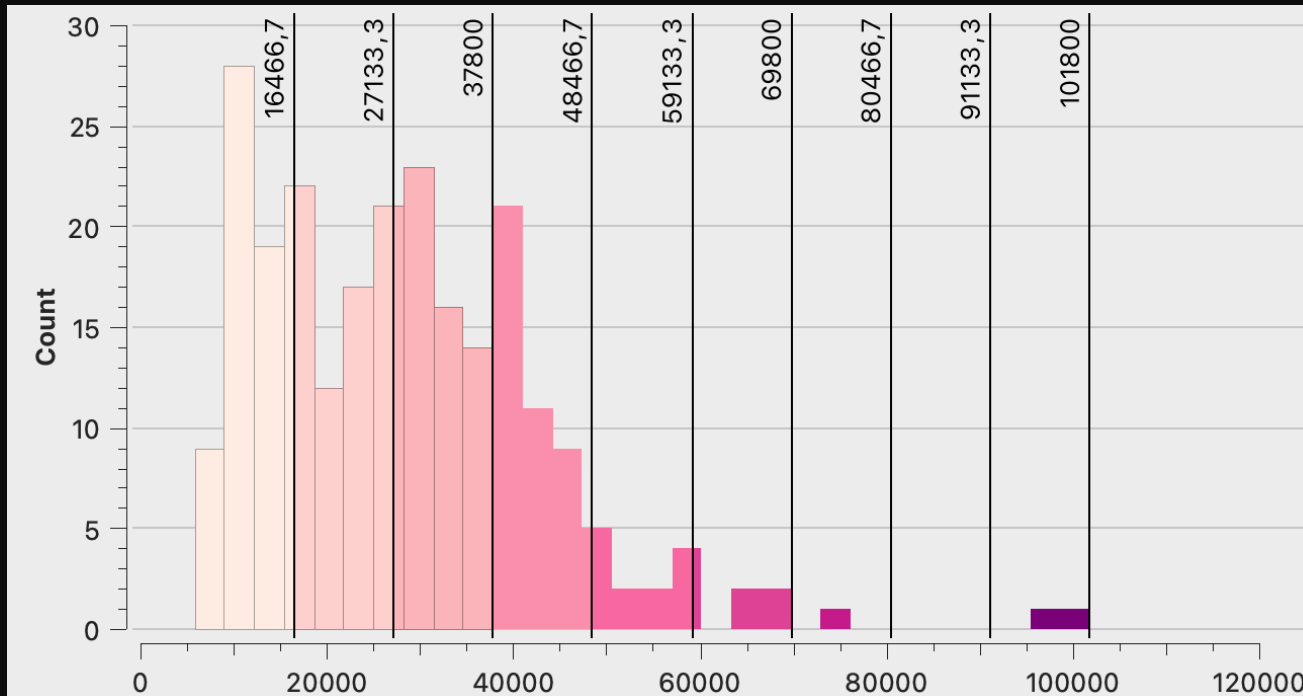
Choropleth map of gross domestic product per capita of EU regions.

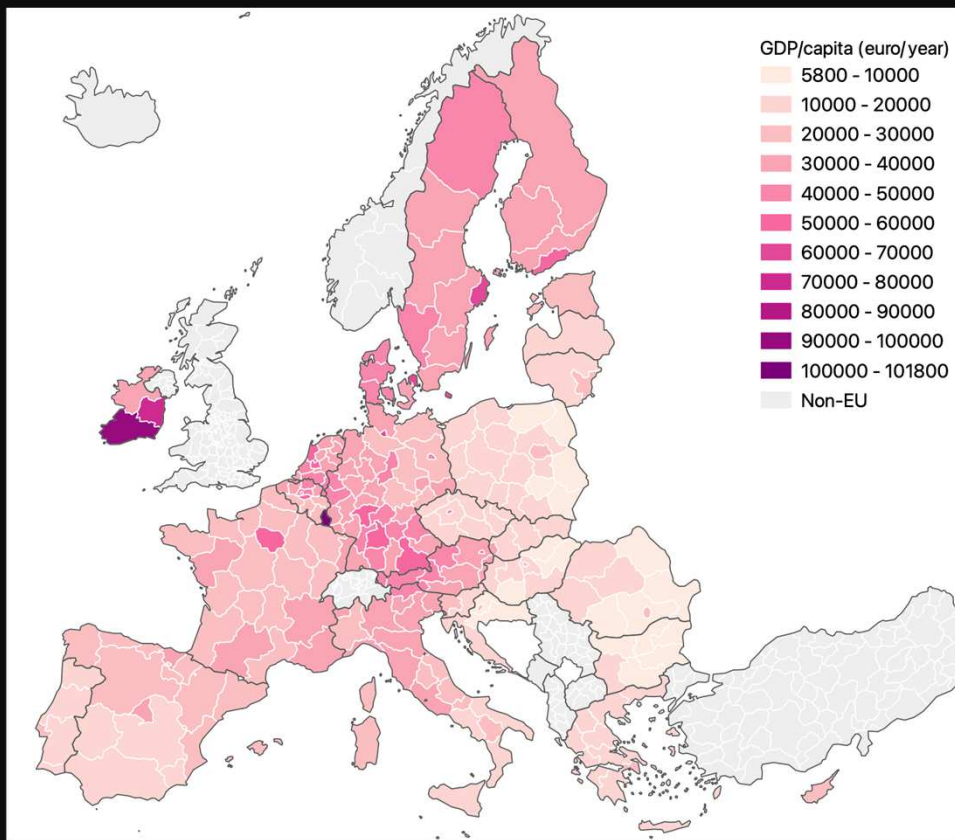
Equal interval

Good for:

Presenting information to a non-technical audience

Easier to interpret, especially if familiar values, like percentages





Pretty Breaks

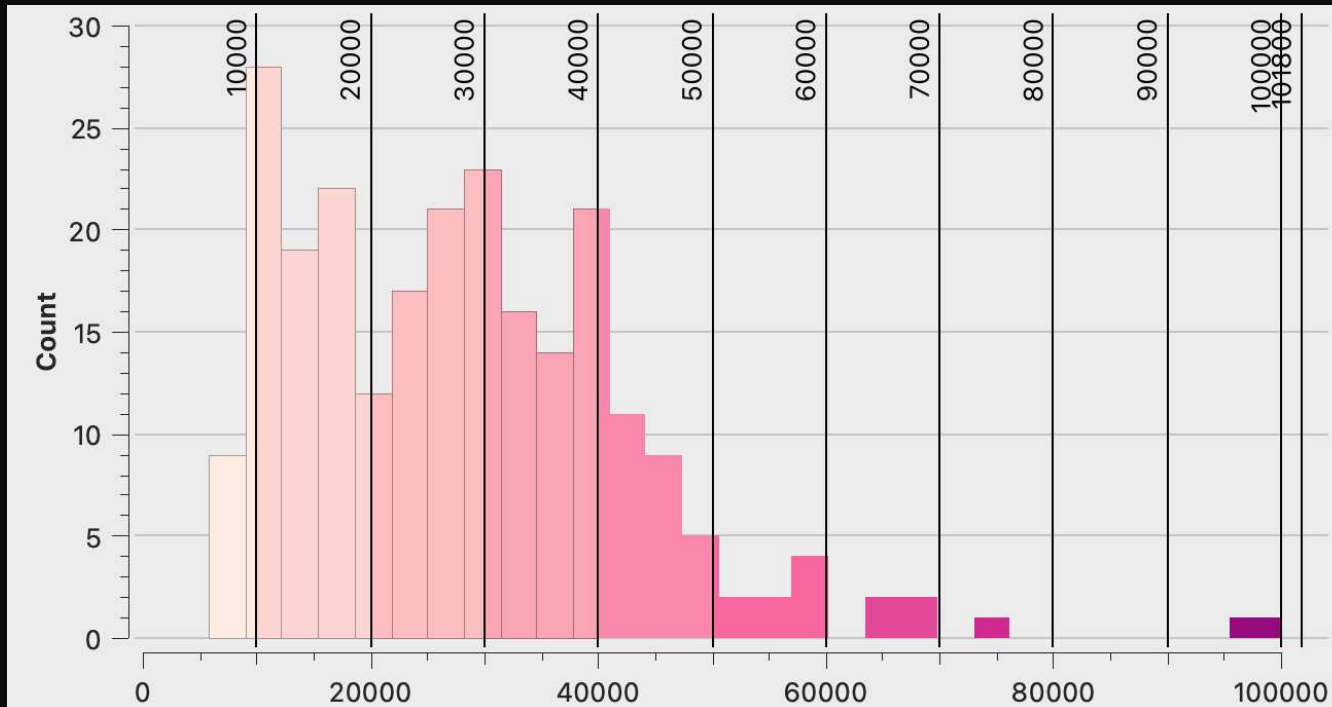
Good for:

Produces neat, rounded class boundaries (e.g., 10, 20, 50, 100).
Easy to interpret and creates readable map legends.

Disadvantages:

Class boundaries may not reflect the natural distribution of the data.

May split natural clusters or combine dissimilar values



Equal count (Quantile)

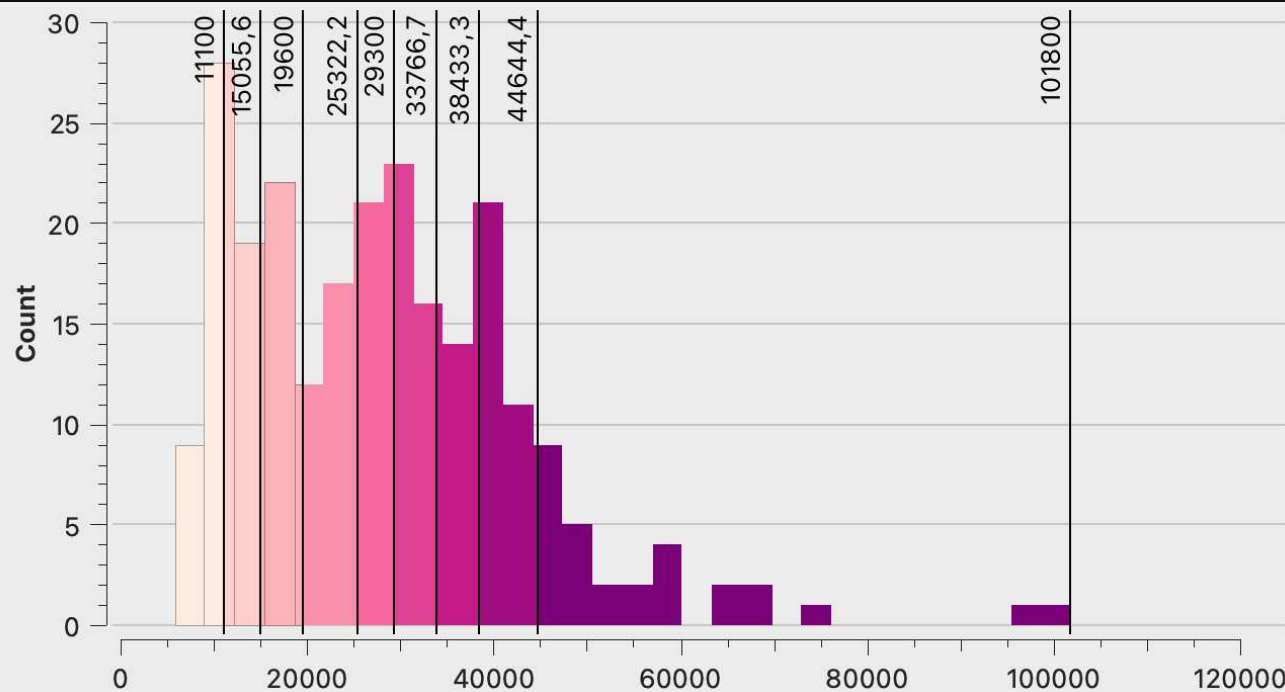
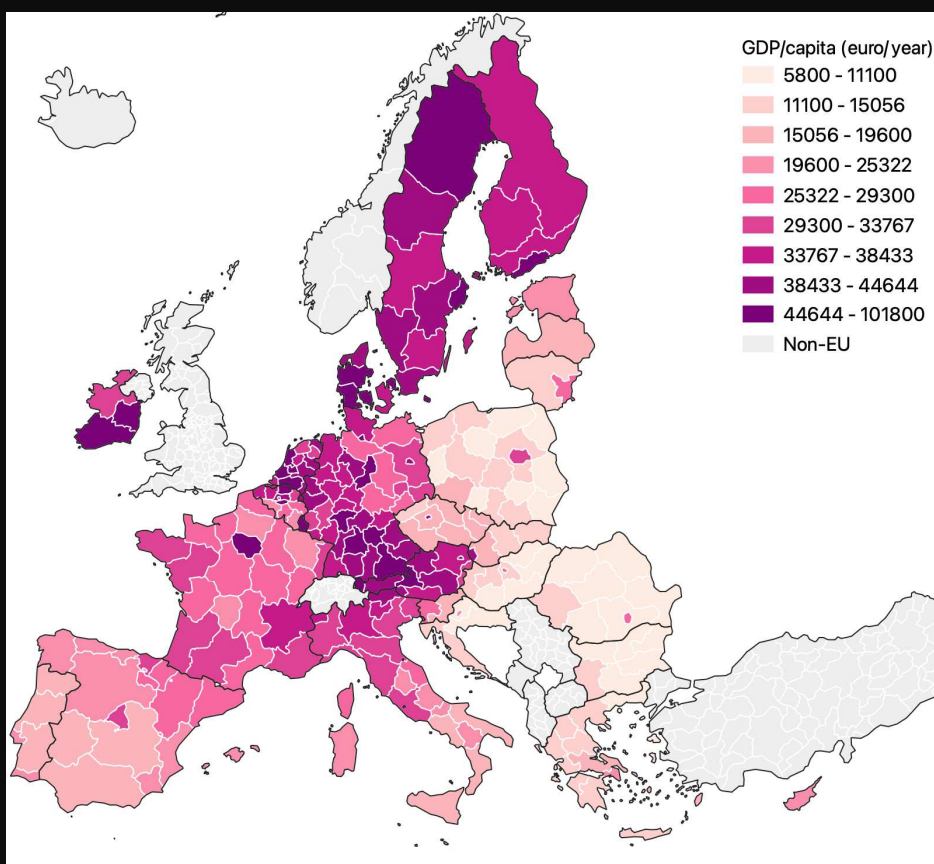
Good for:

Emphasizing relative position e.g. show which districts are in the top 20%, middle 20%, or bottom 20% of the distribution.

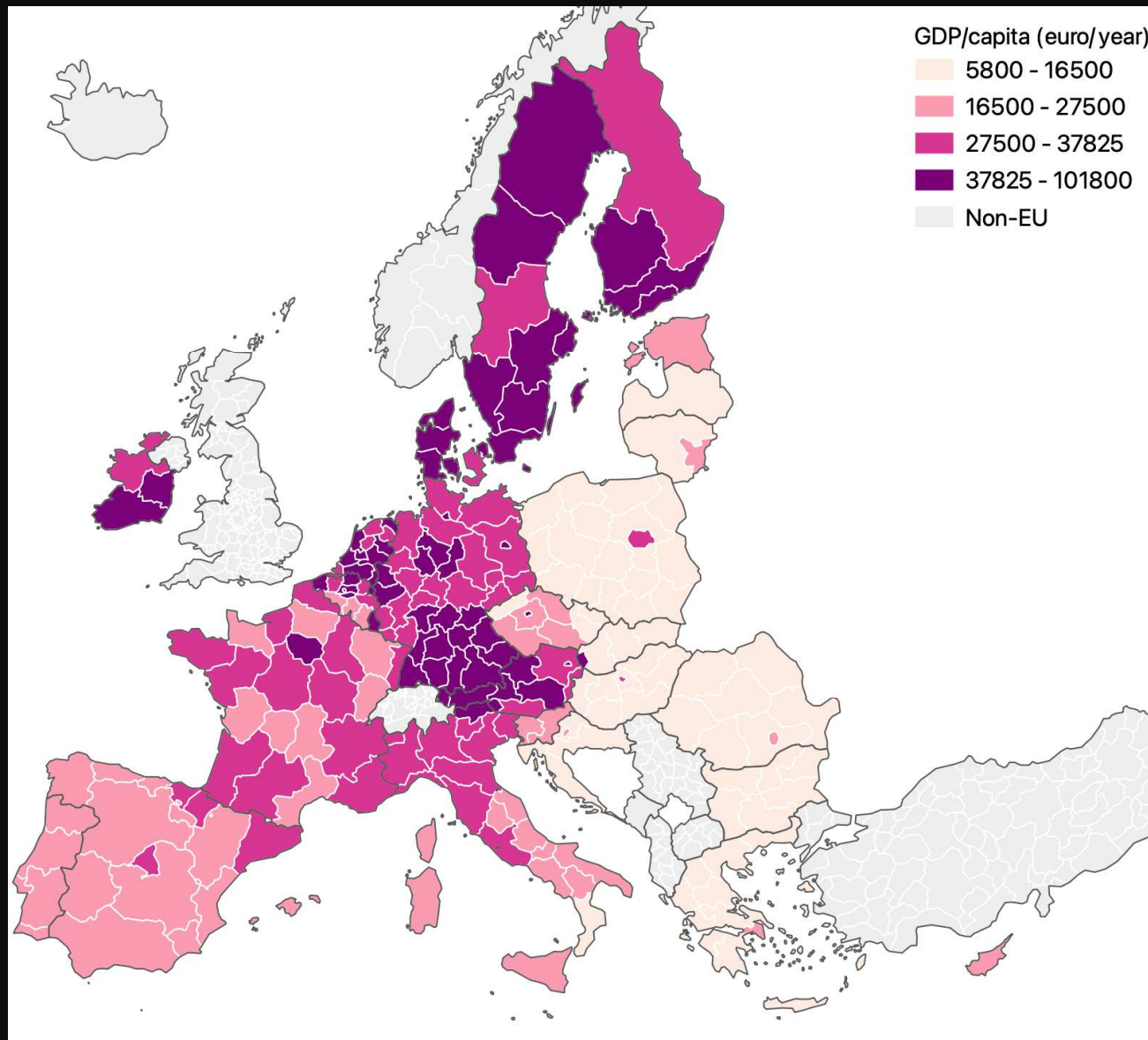
Disadvantages:

Features with similar values may end up in different classes, exaggerating their differences.

Reverse can also happen: wide range of values can end up in same class, minimizing differences



Equal count (Quantile)



Source: Maarten Lambrechts, CC BY
SA 4.0

Standard Deviation

Good for:

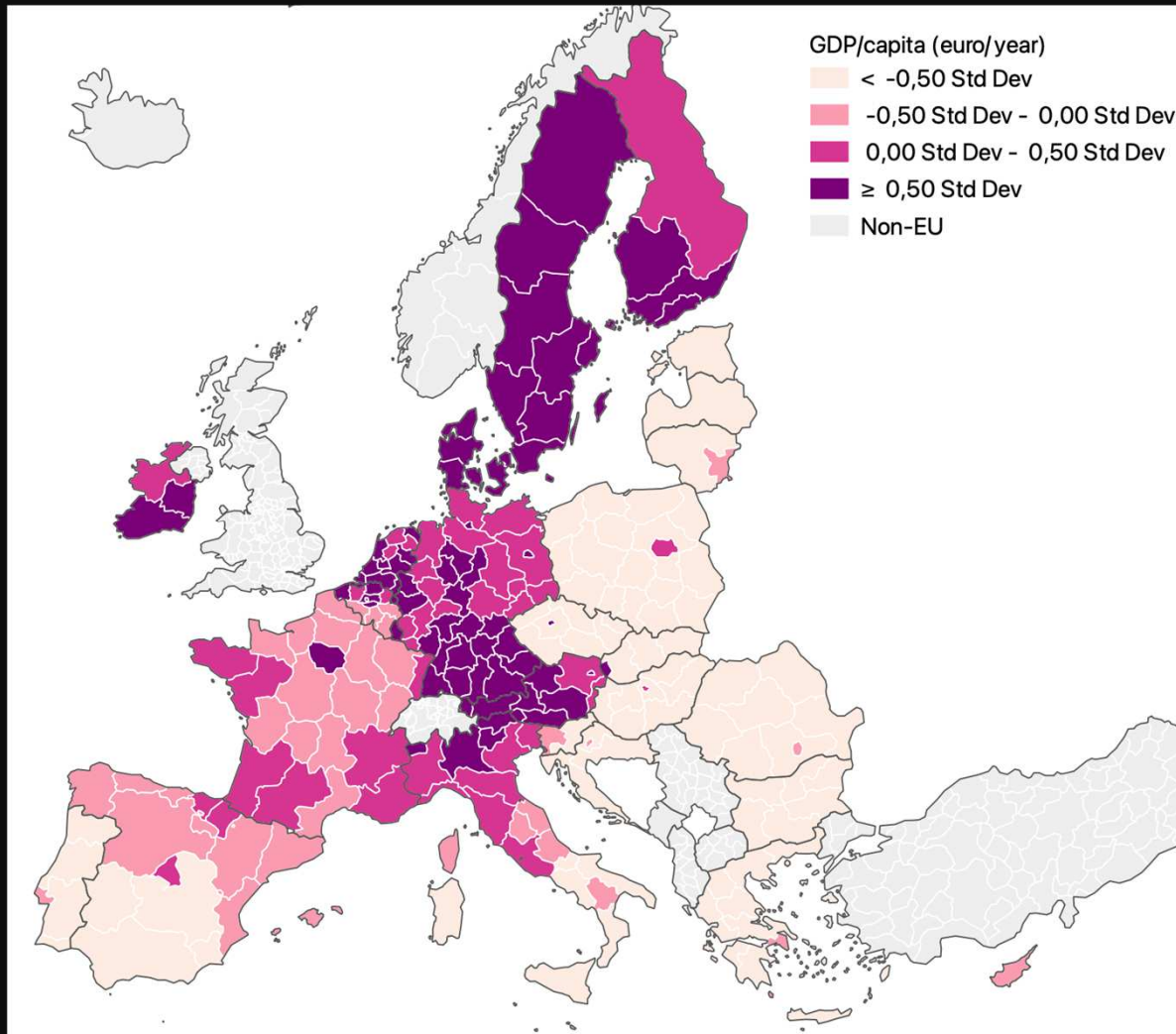
Seeing which features are above or below an average value

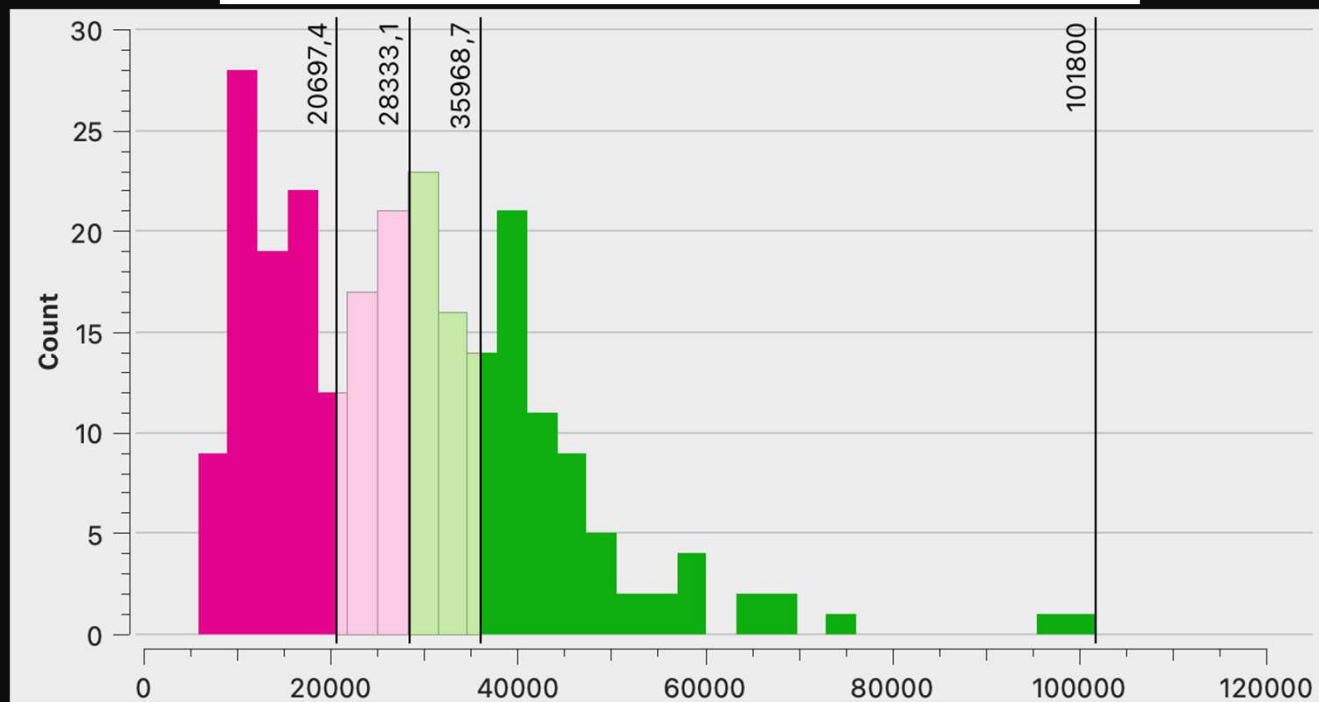
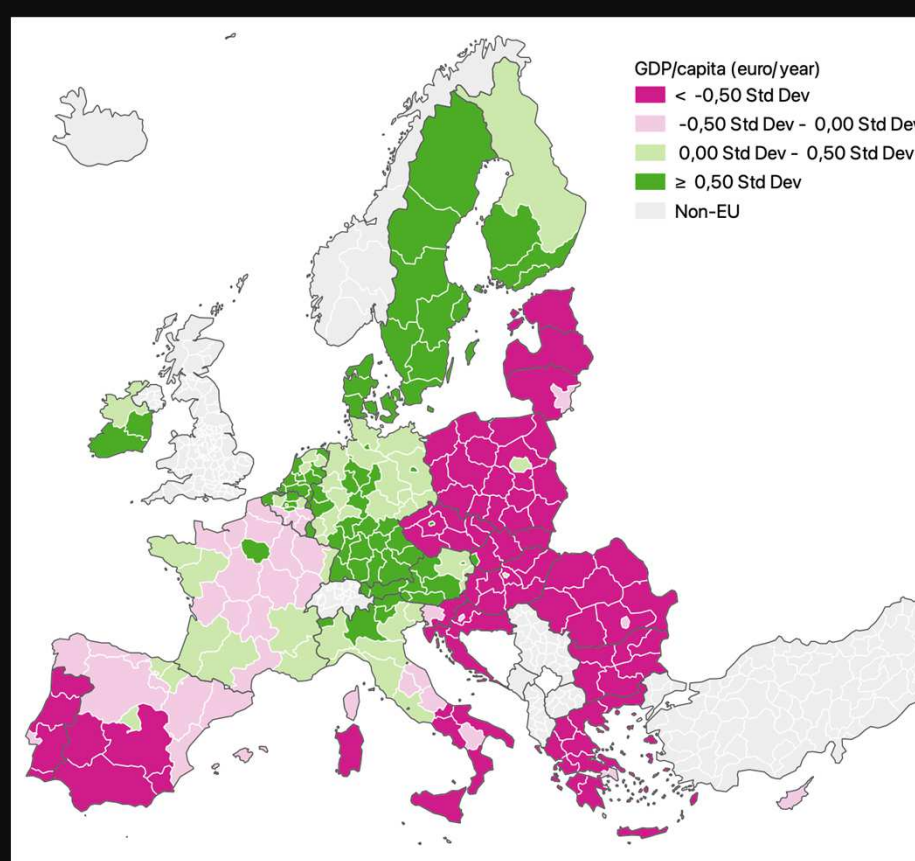
- Data that has a normal distribution

Disadvantages:

Map doesn't show actual values, only how far values are from the mean

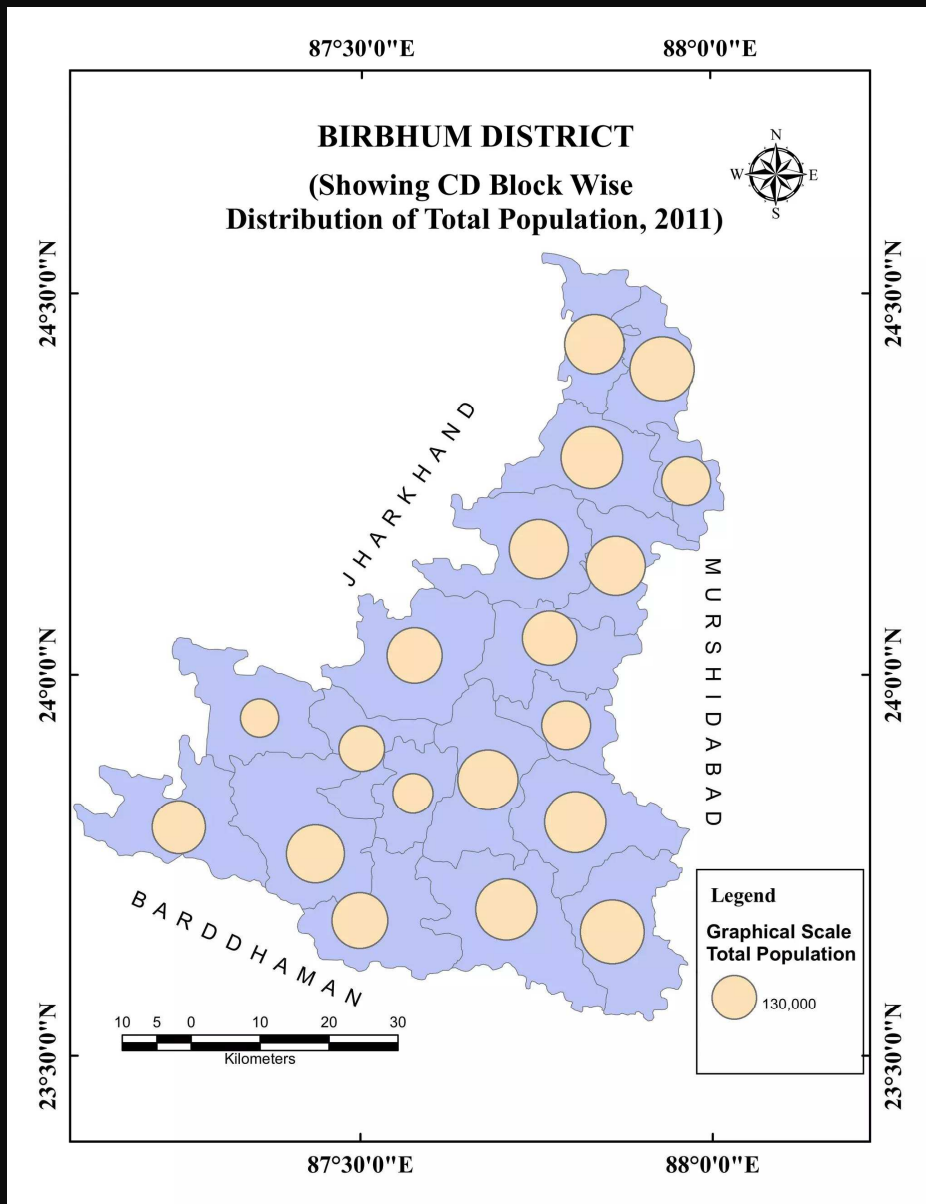
- Very high or low values can skew the mean



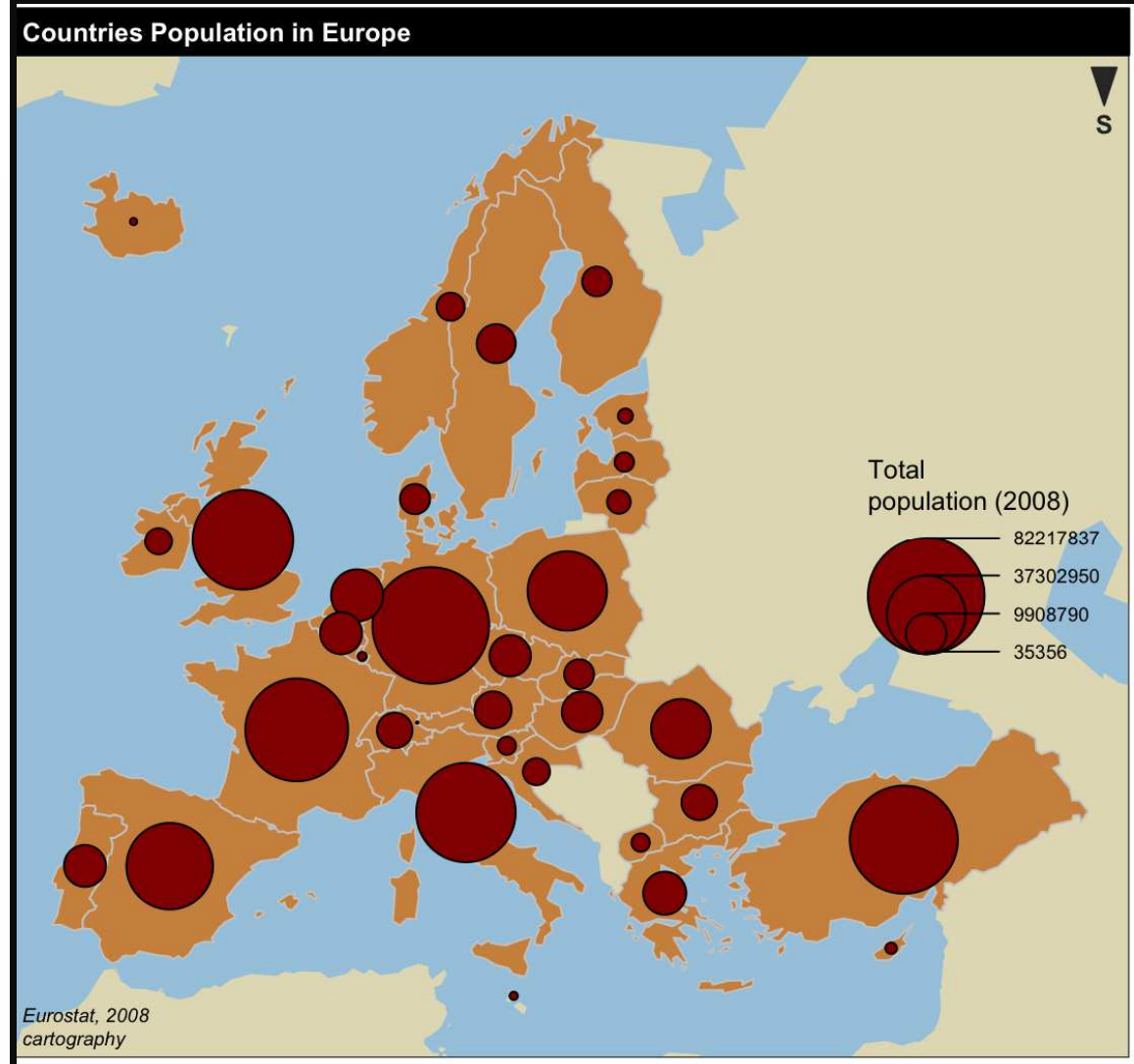


Other thematic representations

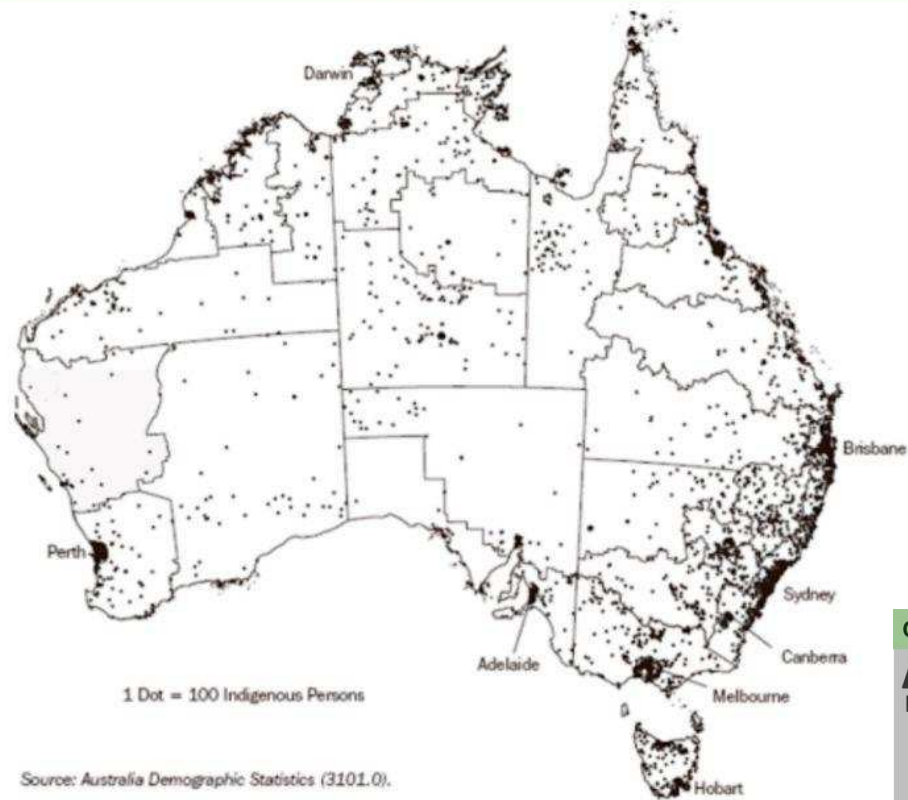
Graduated Symbols



Proportional Symbols

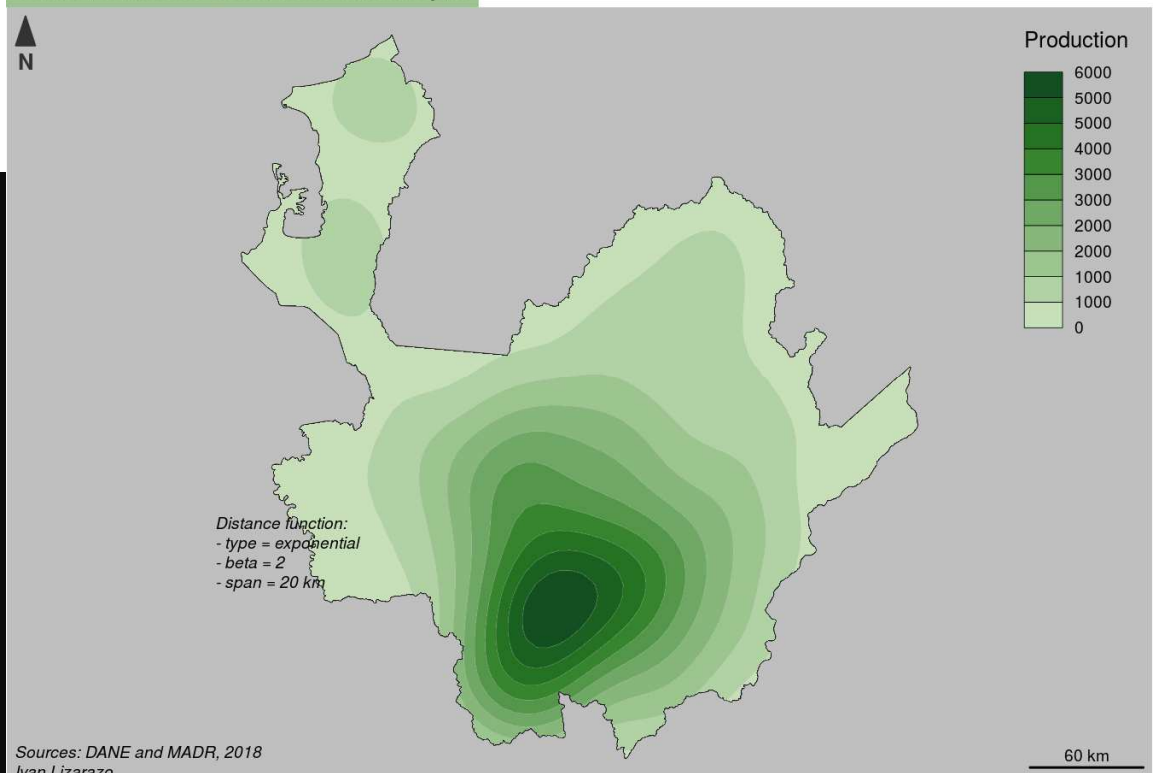


DOT DISTRIBUTION MAP IN GIS



Isopleth Maps

Coffee Production Distribution in Antioquia



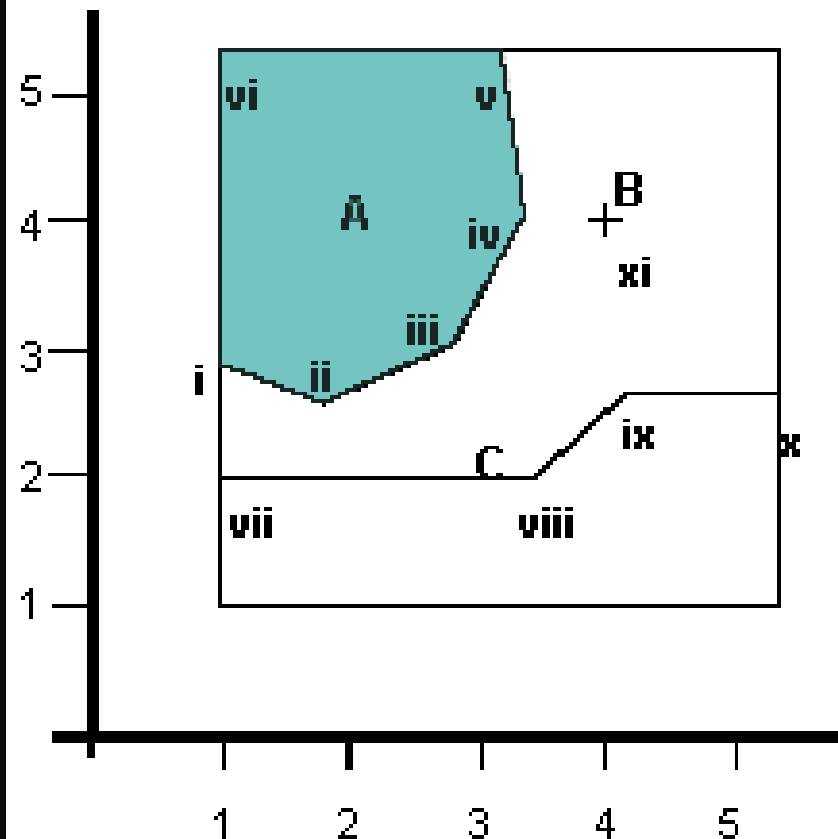
Vector Data Types

There are different models to store and manage vector information. Each of them has different advantages and disadvantages.

Non-Topological
- Spaghetti
Vertex dictionary

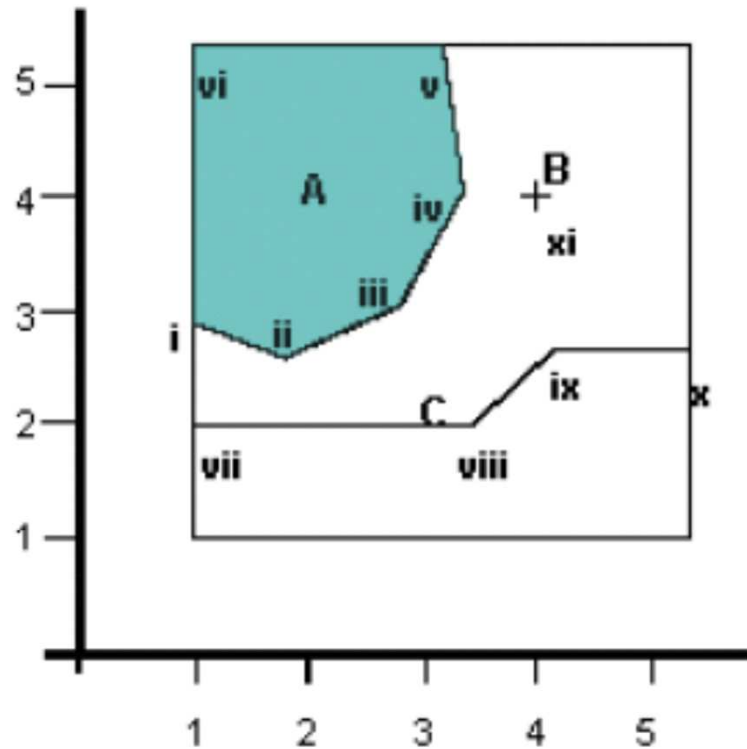
Topological
Dual Independent Map Encoding
(DIME)
Arc / Node

Spaghetti



A, 6 (identifier of polygon and number of vertex)
1, 3 (coordinates of the first vertex)
1.8, 2.6
2.8, 3
3.3, 4
3.2, 5.2
1, 5.2
1, 3 (coordinates of the first vertex again)
B, 1 (identifier of the point and number of vertex)
4, 4
C, 4 (identifier of the line and number of vertex)
1, 2
3.5, 2
4.2, 2.7
5.2, 2.7

Vertex dictionary



	<u>file 1</u>	
vertex	X	Y
i	1	3
ii	1.8	2.6
iii	2.8	3
iv	3.3	4
v	3.2	5.2
vi	1	5.2
vii	1	2
viii	3.5	2
ix	4.2	2.7
x	5.2	2.7
xi	4	4

file 2
polygon A: i, ii, iii, iv, v, vi
point B: xi
line C: vii, viii, ix, x

no duplication, but still this model does not use topology

Entity Relations

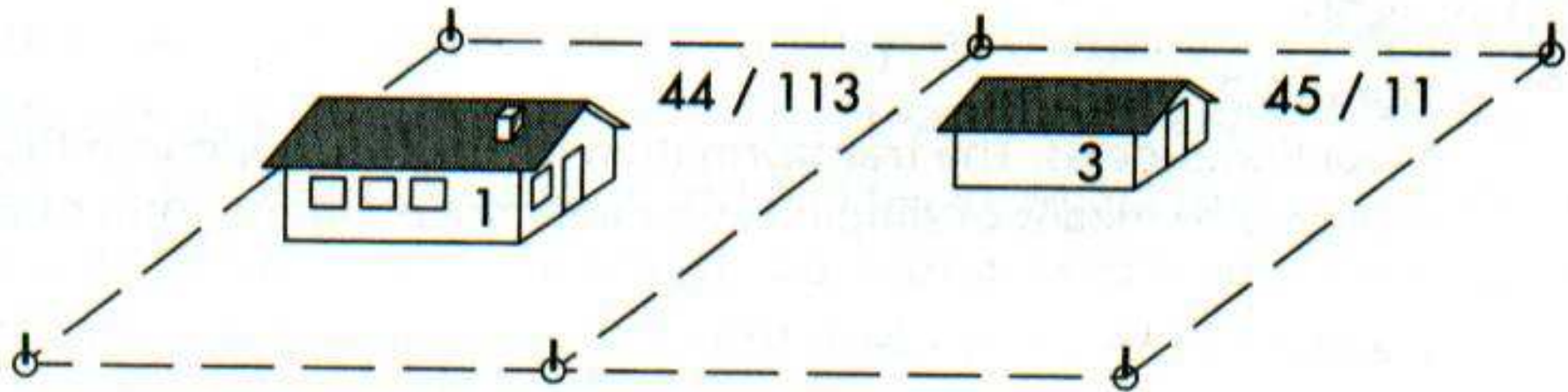


Figure 3.5

The computer cannot see the real world so it is necessary to specify the various relations between entities, such as *belong to*, *comprise*, *are located in/on*, and *border on*.

Topologic data Model

More commonly used in software that implements a full range of operations on vector representations.

Incorporates network relationships along with the coordinate measurements (Chrisman, 1997).

Often referred to as an intelligent data structure because spatial relationships between geographic features are easily derived when using them.

Is the dominant vector data structure currently used in GIS technology

DIME (Dual Independent Map Encoding)

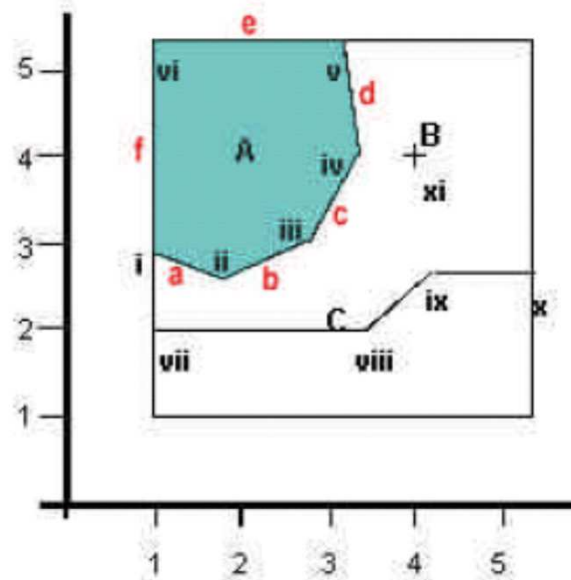
Developed by US Bureau (1968) of the Census for demographic analysis (for storing addresses and urban maps).

Line segments are assumed to be straight When curved lines are needed they are represented as sequence of straight line segments

Each line segment is stored with three essential components; a segment name, node identifiers for the from and to endpoints of the segment and identifiers for the polygons on left and right sides of the segment

The model allows the spatial definitions of map vectors to be stored in non redundant manner, advantageous in defining boundaries of adjacent polygons

DIME



file 1

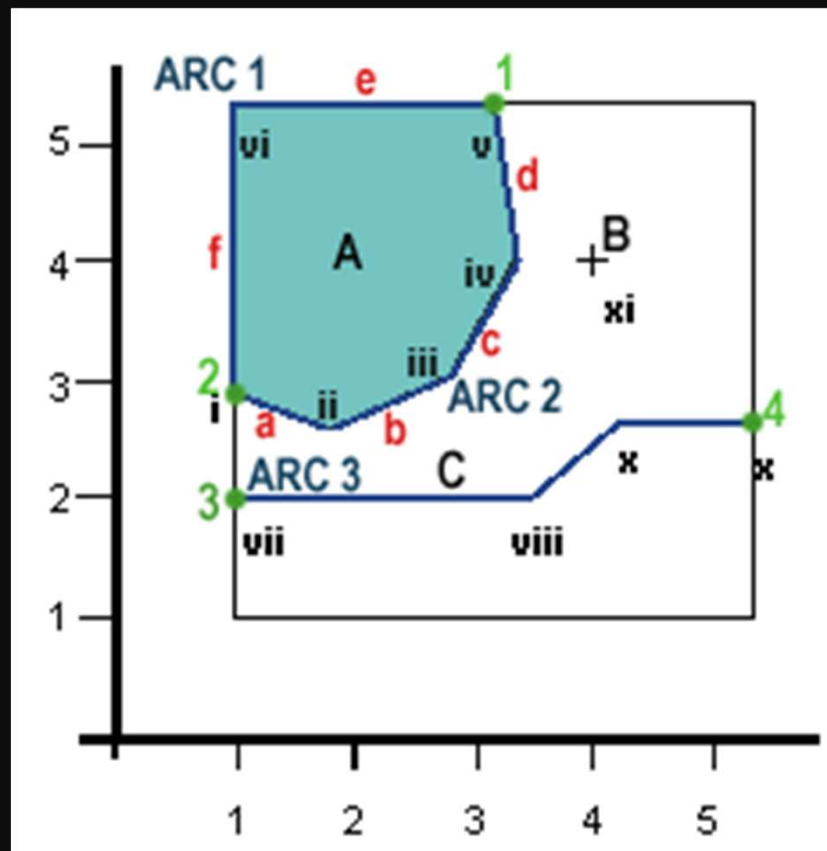
vertex	X	Y
i	1	3
ii	1.8	2.6
iii	2.8	3
iv	3.3	4
v	3.2	5.2
vi	1	5.2
vii	1	2
viii	3.5	2
ix	4.2	2.7
x	5.2	2.7
xi	4	4

segment right polygon left polygon from vertex to vertex

a	external	A	i	ii
b	external	A	ii	iii
c	external	A	iii	iv
d	external	A	iv	v
e	external	A	v	vi
f	external	A	vi	i
g	external	external	vii	viii
h	external	external	viii	ix
i	external	external	ix	x

polygon	segments
A	a, b, c, d, e, f

Arc-Node Structure



File 1. Coordinates of nodes and vertex for all the arcs

ARC	F_node	Vertex	T_node
1	3,2, 5,2	1, 5,2	1,3
2	1,3	.8,2.6 2.8,3 3.3,4	3,2, 5,2
3	1,2	3.5,2 4.2,2.7	5,2,2.7

File 2. Arcs topology

ARC	F_node	T_node	R_poly	L_poly
1	1	2	External	A
2	2	1	A	External
3	3	4	External	External

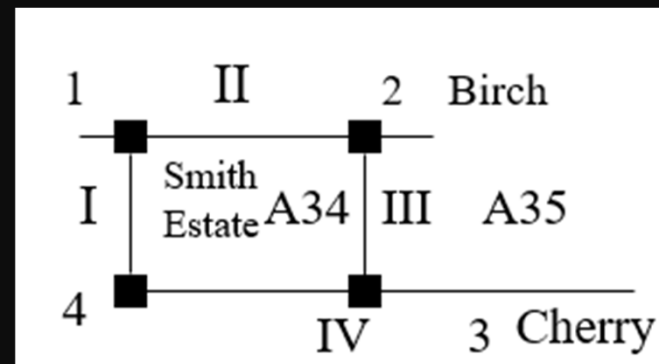
File 3. Polygons topology

Polygon	Arcs
A	1, 2

File 4. Nodes topology

Node	Arcs
1	1,2
2	1,2
3	3
4	4

Arc/Node Structure



Node Table		
Node ID	Easting	Northing
1	126.5	578.1
2	218.6	581.9
3	224.2	470.4
4	129.1	471.9

Arc Table				
Arc ID	From N	To N	L Poly	R Poly
I	4	1		A34
II	1	2		A34
III	2	3	A35	A34
IV	3	4		A34

Polygon Table	
Polygon ID	Arc List
A34	I, II, III, IV
A35	III, VI, VII, XI

Attribute Data

Node Feature Attribute Table			
Node ID	Control	Crosswalk	ADA?
1	light	yes	yes
2	stop	no	no
3	yield	no	no
4	none	yes	no

Arc Feature Attribute Table				
Arc ID	Length	Condition	Lanes	Name
I	106	good	4	
II	92	poor	4	Birch
III	111	fair	2	
IV	95	fair	2	Cherry

Polygon Feature Attribute Table		
Polygon ID	Owner	Address
A34	J. Smith	500 Birch
A35	R. White	200 Main